Trigonometry: Modelling in Degrees

Questions 1–11

Question 1. State the period of each of the following functions:

a) $f(x) = \sin 2x$

b) $f(x) = \cos 6x$

c) $f(x) = \tan x$

d) $f(x) = \sin\left(\frac{1}{2}x\right)$

e) $f(x) = \cos\left(\frac{1}{4}x\right)$

f) $f(x) = \tan(2x - 45)$

g) $f(x) = \sin\left(\frac{360}{7}x\right)$

h) $f(x) = \cos\left(\frac{90}{13}x + \frac{360}{13}\right)$

i) $f(x) = \sin\left(\frac{72}{73}(x - 30)\right)$

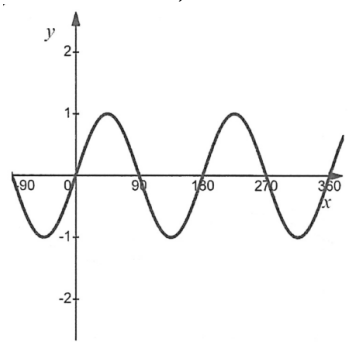
j) $f(x) = \cos(15(x + 7))$

Question 2. In each of the following cases, the curve

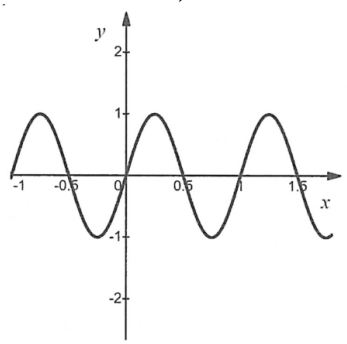
$$y = \sin x$$

has been stretched parallel to the x-axis. State the equation of each curve.

a)



b)

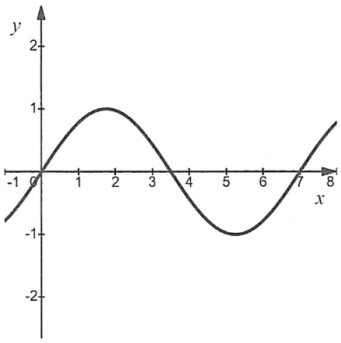


Question 2 (continued). In each of the following cases, the curve

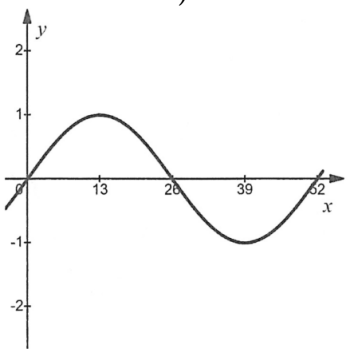
$$y = \sin x$$

has been stretched parallel to the x-axis. State the equation of each curve.

c)



d)

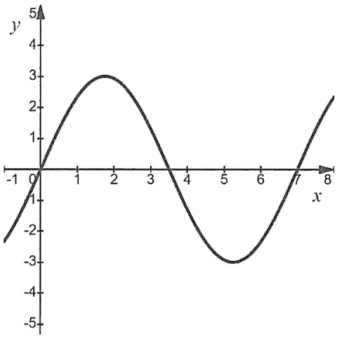


Question 3. All the following graphs are transformations of

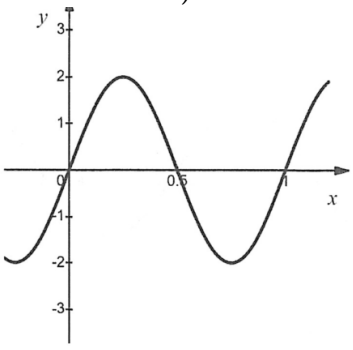
$$y = \sin x.$$

State the equation of each:

a)



b)

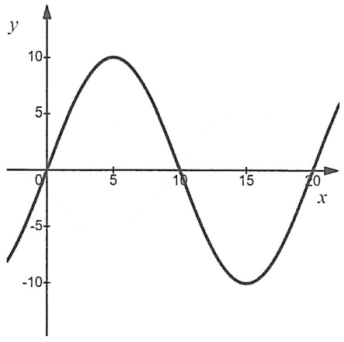


Question 3 (continued). All the following graphs are transformations of

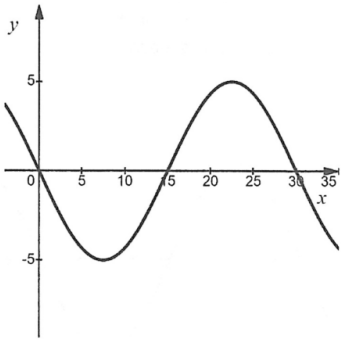
$$y = \sin x.$$

State the equation of each:

c)



d)



Question 4. State the amplitude of each of the following periodic functions:

a) $y = \sin 2x$

b) $y = 2 + \cos(15x + 3)$

c) $y = 40 + 18\sin(12x - 4)$

d) $y = 3 - 5\sin(0.5x + 1.1)$

Question 5. For each of the following functions state the minimum and maximum of the function:

a) $y = \sin 3x$

b) $y = 4 + \cos(12x^\circ - 7)$

c) $y = 25 + 15\sin(30x + 2)$

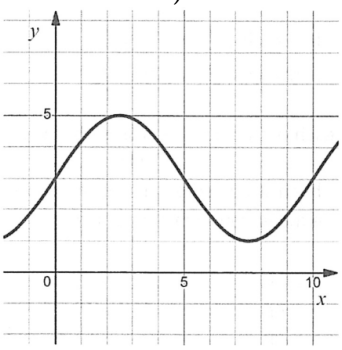
d) $y = 3 - 8\sin(0.5x + 0.2)$

Question 6. All the following graphs are of the form

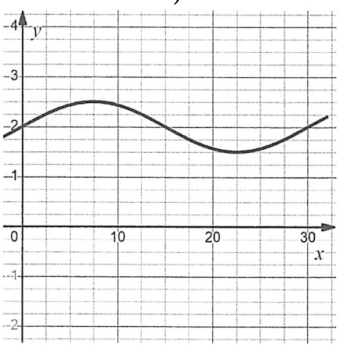
$$y = a + b\sin(cx).$$

In each case, find the equation of the graph:

a)



b)

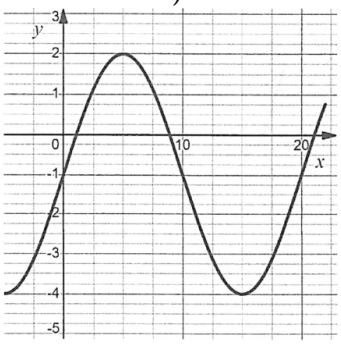


Question 6 (continued). All the following graphs are of the form

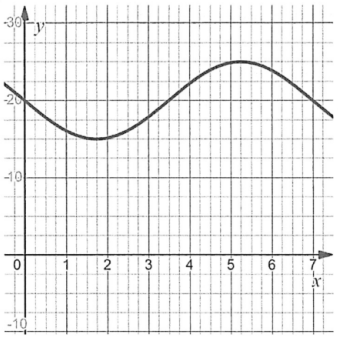
$$y = a + b\sin(cx).$$

In each case, find the equation of the graph:

c)



d)



Question 7. The time of sunrise, y hours since midnight for each day x of a year may be modelled by the equation

$$y = a + b\cos x^\circ.$$

It is known that the latest sunrise is at 8:00 am on January 1st and the earliest is at 4:30 am, six months later.

a) Find the values of a and b.

b) Suggest how the model may be improved.

Question 8. A person's blood pressure, y, oscillates between 120 and 80 with their heartbeat occurring every 1 second.

a) Find a model in the form

$$y = a + b\sin kt^\circ$$

where t is the time that has passed in seconds and a, b, k are constants.

b) Explain why this model will not always be accurate for predicting blood pressure.

Question 9. The size of a population, P, of wild mice is recorded over time, and may be modelled by a formula of the form

$$P = 20 + 6\sin(3t - 10)^\circ, \qquad 0 \leq t \leq 121$$

where t is the number of days since the start of the year. According to the model, find

a) the number of mice in the population after 5 days have passed.

b) the maximum and minimum number of mice in the population.

c) the days on which the maximum and the minimum number of mice in the population occurs.

Question 10. The profits on the sales of ice cream for a shop, £P, may be given by the equation

$$P = 55 - 50\cos\left(\frac{72}{73}(x - 45)\right)^{\circ}$$

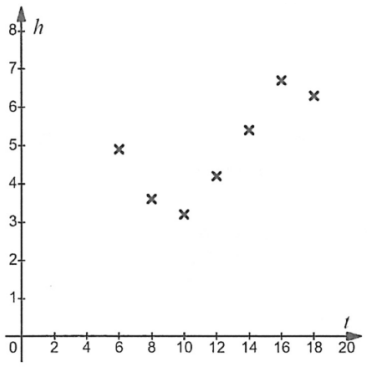
where x is the number of days since the 1st January. According to the model,

a) write down the difference between the largest and smallest values of the profits.

During the 136th day of the year, the rate of change of the profits is at its maximum.

b) Find the next day when the model predicts the rate of change will be the same.

Question 11. Laiyla is monitoring the height of the tide, h metres, in her local harbour. At various points of time, she records the height along with the time t hours after midnight.



Question 11 (continued). Laiyla models the data with the equation

$$h = 5 + 1.7\sin(25t - 330)^\circ$$

where t is the number of hours since midnight.

a) Find the minimum height of the tide predicted by Laiyla's model.

b) Find the maximum number of consecutive hours that Laiyla's model predicts that the height of the tide will exceed 6 metres.

c) Find the second time after midnight at which the tide has a height of 6 metres.

Question 11 (continued). Further along the coast, a second location's tide has the same times for its high and low tides, varying in depth by 3.3 metres. At high tide, it is 2.55 metres less deep than the high tide found by Laiyla's model.

d) Make appropriate refinements to Laiyla's equation to create a model for the height of the tide.